

Credit Card Approval Prediction System – Step-by-Step Development Process

Objective: Develop a machine learning system to predict whether a credit card application should be approved or rejected.

Workflow:

1. Environment Setup
2. Dataset Collection
3. Data Analysis
4. Data Pre-processing
5. Feature Engineering
6. Model Building
7. Model Evaluation
8. Model Selection
9. Application Development
10. Testing & Deployment

Epic 1: Data Collection

Download the credit card approval dataset, store it as CSV, and load it using Python (Pandas).

Epic 2: Visualizing and Analysing the Data

- Import libraries (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn).
- Read and explore the dataset.
- Perform univariate analysis.
- Perform multivariate analysis.
- Generate descriptive statistics.

Epic 3: Data Pre-processing

- Remove duplicates.
- Handle missing values.
- Clean and merge data.
- Perform feature engineering.
- Encode categorical variables.

Epic 4: Model Building

- Split data into training and testing sets.
 - Train Logistic Regression.
 - Train Decision Tree.
 - Train Random Forest.
 - Compare models using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC.
- Select the best model.

Epic 5: Application Building

- Design HTML pages.
- Develop Flask backend and integrate the trained model.
- Test and validate the application.

Deployment

Save the trained model, integrate it into the application, test predictions, and deploy for end users.